CO2 AT 1

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Q. Your organization is migrating an application originally written against a Type 2 (native-API) JDBC driver, which requires vendor client libraries installed on every application server, to a Type 4 (pure Java, thin) driver. Explain the deployment and portability benefit this migration achieves.

Ans:

Aim: To demonstrate the deployment and portability benefits of migrating an application from a Type 2 (Native-API) JDBC driver to a Type 4 (Pure Java/Thin) JDBC driver.

Algorithm:

- Start the Java application.
- Import the JDBC packages.
- Load/register the Type 4 JDBC driver.
- Specify the database URL, username, and password.
- Establish a database connection using Driver Manager.
- Execute a simple SQL query.
- Display the result.
- Close the connection.
- Stop.

Pseudocode:

```
BEGIN
    Import JDBC library
    Load Type 4 JDBC driver
    Set database URL
    Set username
    Set password
    TRY
        Establish connection using Driver Manager
        IF connection is successful THEN
            Display "Database connected successfully"
            Execute SQL query
            Display query result
        ELSE
            Display "Connection failed"
        END IF
        Close connection
    CATCH database exception
        Display error message
    END TRY
END
```

JAVA CODE:

```
import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.ResultSet;

import java.sql.Statement;

public class Type4JDBCExample {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/testdb";

        String username = "root";

        String password = "root";

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, username, password);

            System.out.println("Database connected successfully!");

            System.out.println("Type 4 JDBC Driver is being used.");

            System.out.println("No native client library is required.");

            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery("SELECT 1 AS connection_test");

            while (rs.next()) {

                System.out.println("Query Result: " + rs.getInt("connection_test"));

            }

            rs.close();

            stmt.close();

            con.close();

            System.out.println("Connection closed.");

        } catch (Exception e) {

            System.out.println("Database connection failed.");

            System.out.println("Error: " + e.getMessage());

        }

    } }
```

OUTPUT
Database connected successfully!

Type 4 JDBC Driver is being used.

No native client library is required.

Query Result: 1

Connection closed.